DRAWVERIFY: VALIDATING SKETCHED ROBOT INSTRUCTIONS BEFORE EXECUTION

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ABSTRACT

Circle-and-arrow sketches tell a robot *which* object to move and *where* without naming anything, but existing sketch-conditioned systems execute the drawing unchecked. We study **sketch-prompt validation**—deciding *before* any action whether a drawn instruction is correct, consistent across circle, arrow, and caption, and sufficient for a unique action, and saying why not—as an explicit pre-execution gate. A pilot by team members built a simulator-grounded generator over LIBERO (8,522 labelled prompts), collected 275 human sketches, and ran a 17-participant in-the-loop study: generated prompts rank validators like human sketches do but inflate AUROC by $+0.13$; a LoRA-adapted Qwen3-VL-4B staged over 82 human sketches reaches 0.808 AUROC, on par with 4-shot Gemini-3.8-Flash; yet live gating gave no detectable improvement, with in-loop precision 0.38 and no diagnostic in the rejection. We propose to (H1) match generated strokes to human statistics and audit geometry-only ceilings, (H2) *generate* a per-channel verdict and rejection reason, (H3) calibrate the operating point with a conformal false-accept guarantee, and (H4) test channel-naming feedback and a one-retry cap in a follow-up study.

1 PROBLEM AND MOTIVATION

Language is ambiguous in cluttered scenes with duplicate objects and goal images are over-specified (Sundaresan et al., 2024); a circle-and-arrow sketch with a name-free caption ("move this there") is a persistent deictic act (Sauppé & Mutlu, 2014; Kranstedt et al., 2006). Recent systems condition robot policies on such sketches (Sundaresan et al., 2024; Gu et al., 2024; Li et al., 2025a;b) and all silently assume the drawing is *valid*. Yet circles miss or enclose a distractor, arrows point to the wrong container or detach from the circle, captions contradict sketches, and channels that look fine alone can jointly disagree; executing such a sketch commits the robot to a wrong action.

Problem (Track B). *Can a model decide, before any action, whether a drawn instruction is safe to execute—and say why not?* We formulate **sketch-prompt validation** as a decision over image, circle, arrow, and caption requiring per-channel correctness, cross-channel consistency, and sufficiency

*CS707 Group Project Proposal, Track B (novel task formulation).

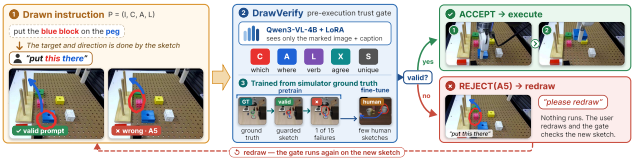


Figure 1: The pilot validator as a pre-execution trust gate (DrawVerify pilot, 2026): a LoRA-adapted VLM trained on simulator ground truth judges the marked image and caption; ACCEPT executes, REJECT asks the user to redraw.

(Section 3.1), plus a generated reason; no existing task captures it (Section 2). We address it with generative VLMs as prompted judges and as LoRA-tuned validators (Figure 1).

Pilot evidence and gap. A pilot by team members (DrawVerify pilot, 2026) built DrawVerify, a simulator-grounded generator over LIBERO (Liu et al., 2023a): 8,522 labelled prompts (31.8% valid) from 800 demonstrations over 40 tasks, 15 of 18 corruption modes plus five grounding configurations, 275 human sketches (82 train / 193 test), and 326 sketches from a 17-participant in-the-loop study. Findings (Appendix B, Figure 3): (i) generated prompts order seven API validators as human sketches do (Spearman $\rho = +0.93$) but inflate AUROC unevenly ($+0.13$ mean, -0.06 to $+0.25$); (ii) Qwen3-VL-4B (Qwen Team, 2025) with LoRA (Hu et al., 2022) trained on generated prompts alone transfers poorly (0.565 AUROC on human sketches) but staged over 82 human sketches reaches 0.808, matching 4-shot Gemini-3.8-Flash (0.857, n.s.); (iii) live gating gave *no detectable improvement*: in-loop precision 0.382, third attempts broke more than they repaired, and the rejection carried no diagnostic. The pilot never measured the human-likeness of generated strokes, never tested a per-channel verdict, fixed its threshold by a single recall target (0.85 recall on the 82 human training sketches) with no finite-sample guarantee, and gave neither gated arm a diagnostic in the rejection. This project delivers those deltas.

Hypotheses. H1 (hardening and transfer): trained on the hardened tier alone, the validator reaches $\geq$0.70 AUROC on the 193 human test sketches (v1-only: 0.565), while a geometry-only classifier on that tier stays $\leq$0.60 AUROC on an untouched audit split. **H2 (structured supervision):** supervising the five predicates as auxiliary generative targets lifts recall on directional failures of the human test split from 0.53 to $\geq$0.70 at no loss in R_{val}, and generated reasons name the violated predicate with $\geq$80% faithfulness on generated negatives and $\geq$70% on human ones. **H3 (calibrated gate):** with $\hat{\tau}$ from Eq. (2), false-accept rate (FAR) on held-out human sketches is $\leq \alpha = 0.10$ in every seed, and at the pilot's recall (0.786) precision on ungated study sketches beats its 0.458 by $\geq$10 points. **H4 (feedback):** in a follow-up study ($\geq$20 participants), a gate that names the failing channel under a one-retry cap raises the validity of the finally sent instruction over the bare gate (pilot: 0.795) by $\geq$17 points, the pilot's minimum detectable effect.

2 Related Work

Sketch- and visual-prompt-based instruction. RT-Sketch and RT-Trajectory condition policies on goal or trajectory sketches (Sundaresan et al., 2024; Gu et al., 2024), RoVI decodes an arrow/circle language with a VLM (Li et al., 2025b), CrayonRobo *tolerates* noisy prompts but does not classify them (Li et al., 2025a), and Action-Sketcher, AnyUser and visual-prompting methods (Tan et al., 2026; Yang et al., 2026; Nasiriany et al., 2024; Yang et al., 2023; Liu et al., 2024; Yuan et al., 2024; Cai et al., 2024) translate or annotate sketches for a VLM. All assume the prompt is valid; *we extend* them by making validity the primary decision.

Verification and safety gating. KnowNo asks for help via conformal prediction (Ren et al., 2023); VerifyLLM and VeriGraph verify plans against temporal logic or scene graphs (Grigorev et al., 2025; Ekpo et al., 2026); DoReMi and REFLECT detect or explain execution failures (Guo et al., 2024; Liu et al., 2023b). These verify *plans* or *executions*; *we contrast* by verifying the instruction itself, an earlier gate, with a conformal threshold (Angelopoulos et al., 2024; Geifman & El-Yaniv, 2017). Shortcut audits (Gururangan et al., 2018; Geirhos et al., 2020; Swayamdipta et al., 2020) filter fixed datasets; *we control the generator*.

3 Task Formulation and Method Sketch

3.1 Formulation

A sketch prompt is $P = (I, C, A, L)$: RGB observation I, circle C (centre, radius) denoting *which* object, arrow A (tail, head) denoting *where* it goes, and a name-free caption L. From simulator ground truth we define predicates v_C (the circle uniquely selects the target), v_A (correct bearing and magnitude, anchored on the circled object, ending in the intended region), v_L (caption verb and implicit referents match the action), cross-modal consistency v_X (all channels describe the same object, destination, and predicate), and sufficiency v_S (a unique executable action). A prompt is valid iff $V(P) = v_C(P) \wedge v_A(P) \wedge v_L(P) \wedge v_X(P) \wedge v_S(P)$; the task is to learn $f_\theta(P) \in$

$\{\text{ACCEPT}, \text{REJECT}\}$ predicting $V(P)$, with a reason on REJECT. A prompt can pass every per-channel predicate yet fail v_X or v_S, so validation is not a set of independent checks.

3.2 Simulator-grounded data generation and shortcut hardening

Pilot pipeline (Figure 1; Appendix Figure 2, top). From LIBERO's object poses, region geometry, and gripper trajectories we extract the correct circle target and arrow endpoints for every frame with no human labelling. Valid prompts get *guarded* human-imprecision augmentation (jitter, non-circular strokes, hand-like shafts), with in-point/out-point guards keeping the target inside the circle and distractors outside. Negatives apply one controlled corruption from four families (15 of 18 modes; Appendix A) plus five grounding configurations, so every negative has a known mode, severity, and violated predicate.

New: shortcut hardening (H1). Let $g(P)$ be the raw geometry vector (circle centre, radius, eccentricity; arrow endpoints, length, bearing, tail offset). (i) *Distribution matching to humans*: the pilot never compared generated and human strokes; we estimate the marginals of g on the 82 human training sketches and importance-re-sample valid imprecision and corruption severities to match them as far as labels allow, guard-crossing modes (near-miss, wrong magnitude) drawing severities from the mirrored distribution just beyond the guard. (ii) *Adversarial filtering* (Le Bras et al., 2020) on medium/hard modes: we repeatedly train geometry-only gradient-boosted classifiers (Ke et al., 2017) on random folds and remove the examples held-out classifiers predict most confidently until geometry-only AUROC on an *untouched* audit split is ≤ 0.60 or retention falls below 60%. We also report a *scene-blind* ceiling (sketches on a blank canvas).

3.3 Structured validator and conformal inference

Model. Qwen3-VL-4B (and 2B) with LoRA (rank 32, as in the pilot) takes the image with the rendered sketch, the caption, and a fixed prompt and generates a *structured verdict*: five predicate questions q_k, $k \in \{C, A, L, X, S\}$, answered sequentially with earlier answers in context (teacher-forced in training), then the ACCEPT/REJECT question q_V, then, on REJECT, a one-sentence reason naming the failing channel. Training keeps the pilot's staged recipe (generated tier, then the 82 human sketches). Predicate labels are free: each generated negative violates a known subset and human negatives carry an annotator failure code.

Objective. Each yes/no answer is scored by forced choice over its two answer tokens, $p_\theta(\text{yes} \mid P, q_k) = \sigma(z_{\text{yes}} - z_{\text{no}})$; the reason uses token-level cross-entropy ℓ_R on REJECT examples against paraphrased template reasons per mode. We minimise

$$\mathcal{L}(\theta) = \ell_V(\theta) + \lambda \sum_{k \in \{C,A,L,X,S\}} \ell_k(\theta) + \mu\, \ell_R(\theta), \qquad \ell_k(\theta) = -\log p_\theta\big(y_k \mid P, q_k\big), \tag{1}$$

with λ, μ tuned on validation; $\lambda = \mu = 0$ recovers the pilot's answer-only validator (ablation A1).

Inference with a finite-sample false-accept guarantee. Let $s(P) = p_\theta(\text{ACCEPT} \mid P, q_V)$, read from the VALID/WRONG token probabilities as in the pilot; we ACCEPT iff $s(P) \geq \hat{\tau}$. False accepts are the safety-critical error, so we bound their rate by conformal risk control (Angelopoulos et al., 2024): on the n *invalid* calibration sketches let $\ell_i(\tau) = \mathbb{1}[s(P_i) \geq \tau]$ and

$$\hat{\tau} = \inf\big\{\tau : \tfrac{1}{n+1}\textstyle\sum_{i=1}^{n} \ell_i(\tau) + \tfrac{1}{n+1} \leq \alpha\big\}. \tag{2}$$

If calibration and test prompts are exchangeable, $\mathbb{E}[\text{FAR}(\hat{\tau})] \leq \alpha$; with $n = 36$ invalid calibration sketches $\alpha \geq 1/(n{+}1)$, hence $\alpha = 0.10$. We report the induced false-reject rate, precision at matched recall, and coverage across seeds, tiers, and study sketches (where exchangeability is *not* guaranteed).

4 Datasets, Metrics, Baselines, and Experimental Plan

Datasets and licences. (1) *LIBERO* (Liu et al., 2023a) (MIT): 800 demonstrations, 40 tasks. (2) *DrawVerify v1* (pilot): 8,522 prompts—6,463 / 900 / 1,159 train / val / test, demo-disjoint, 68.2% invalid (Appendix A)—with image, coordinates, caption, label, and mode. (3) *v2 (hardened)*: same

splits after matching and filtering ($\geq$60% retention expected) plus the untouched audit split. (4) *Human sketches* (pilot; Appendix B): the 275-sketch corpus (157 valid / 118 invalid, task-disjoint 82 train / 193 test, annotator failure codes) and the 326 study sketches (88 ungated); the 82 training sketches double as calibration set. (5) *Follow-up study* (new; IRB exemption first): $\geq$20 adult volunteers, 21 trials each in three counterbalanced blocks (bare gate; channel-naming gate; with one-retry cap), with the pilot's interface and blind labelling deck. Data and adapters: CC BY 4.0; code: MIT.

Metrics. AUROC (invalid as positive class), accuracy, precision and recall on invalid prompts, R_{val} (recall on valid), MCC; FAR, FRR and precision at $\hat{\tau}$; recall per failure family and mode; geometry-only and scene-blind ceilings per tier; ranking agreement ρ with human sets; reason faithfulness against the ground-truth predicate or annotator failure code (string match, Qwen3-8B judge); study outcomes as in the pilot: validity of the finally sent instruction, attempts per trial, fix/break counts (Wilson intervals).

Baselines. (i) Majority class. (ii) The pilot's seven API models (Gemini-3.8-Flash, GPT-5-mini, GPT-4.1-mini, Gemini-2.5-Flash, Claude-Haiku-4.5, GPT-5-nano, Claude-3-Haiku) under the identical 4-shot prompt, cached outputs reused. (iii) Zero-shot Qwen3-VL-2B/4B. (iv) Geometry-only LightGBM fits and the scene-blind validator as ceilings. (v) A fine-tuned discriminative model (ConvNeXt-T, late fusion of $g(P)$ and a caption embedding). (vi) The pilot's staged validators, rationale (0.808 AUROC) and answer-only (0.841). *Ours:* structured validator with the conformal gate.

Ablations ($\geq$3), sensitivity, statistics. A1: structured vs. answer-only vs. free rationale ($\lambda = \mu = 0$; H2). A2: v2 vs. v1, with and without the human stage (H1). A3: modality (no caption; coordinates-as-text; scene-blind). A4: no guarded augmentation. A5: calibration set 40 vs. 82 and $\alpha \in \{0.05, 0.10, 0.20\}$ (H3). A6: backbone Qwen3-VL-2B/4B, Qwen2-VL-2B. Sensitivity: training-set size, LoRA rank, corruption severity. 3 seeds (mean$\pm$std); paired bootstrap (2,000 resamples) for AUROC differences, McNemar's test (Dietterich, 1998) on seed-matched decisions, Williams' test for dependent ρ, minimum detectable effects per slice and for the study.

5 Risks, Resources, and Team Plan

Risks and mitigations. Hardening over-removes or still leaks: per-mode retention curves, removal capped at 40%, both tiers released. **Small human sets:** 36 invalid calibration and 82 invalid test sketches; every human-set claim carries bootstrap CIs. **Study power:** the pilot's minimum detectable effect was 17 points with 12 participants; we recruit $\geq$20, within-participant, and replay logged attempts under the one-retry cap. **API drift, cost, compute:** pinned versions, cached pilot outputs; $\approx$2 GPU-h per Qwen3-VL-4B run (estimated from the pilot's 2-epoch stage), $\approx$45 runs $\approx$100 GPU-h on one A100/4090, $\approx$USD 100–150 of API calls. **Schedule:** go/no-go 25 Oct—if the v2 ceiling is still ≥ 0.75 at 60% retention we report the best partial tier and shift effort to H2–H4. **Over-claiming safety:** the gate reduces, not eliminates, unsafe executions; nothing is executed on a robot.

Roles and timeline (2026). *Member 1 (data):* generator v2, matching, audit, release. *Member 2 (modelling):* structured objective, reasons, gate, LoRA training, ablations. *Member 3 (evaluation):* API baselines, statistics, follow-up study, ethics. *1–11 Oct:* IRB, recruitment, feedback interface. *12–25 Oct:* generator v2, structured validator and gate, A1–A3; go/no-go. *26 Oct–8 Nov:* follow-up study (H4), A4–A6, statistics. *9–12 Nov:* final presentation ($\geq$90% of results); *13–27 Nov:* final report and artifacts.

Ethics and reproducibility statements

Training data are simulator renders with no people or personal data; both sketch studies use consenting adult volunteers and store only drawings, prompts, and anonymous ratings. A false accept could cause a wrong physical action, so we report false-accept rates with confidence intervals and recommend human confirmation at deployment. Licences: LIBERO (MIT), Qwen (Apache-2.0), API terms; we release data and adapters under CC BY 4.0 and code under MIT, with seeds, cached API outputs, both tiers with checksums, and study and compute logs.

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A Failure Taxonomy and Dataset Composition

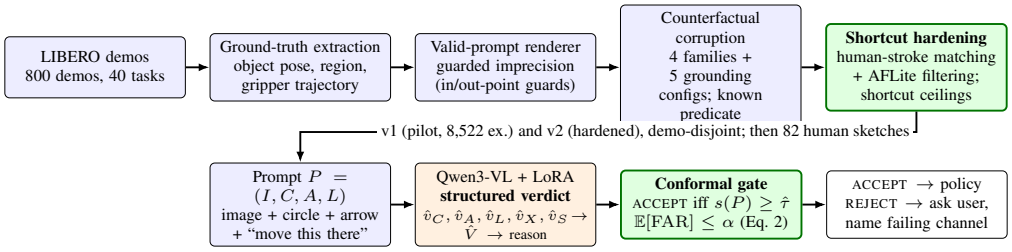


Figure 2: Method sketch (referenced from Section 3.2). *Top:* simulator-grounded data generation (green = new). *Bottom:* the validator generates a structured verdict and a reason; a conformal threshold yields an execute/ask decision.

Table 1: Failure taxonomy of the pilot release. Families A–D define 18 corruption modes, of which 15 are instantiated (Cov. ✓); C2, D1, D2 are unreachable under name-free captions (N/A) because such a caption carries no noun to corrupt. The Grounding family is counted separately. Mode IDs are the generator's keys and are non-contiguous.

Family	ID	Mode and generation mechanism	Cov.
Referential (circle / "which"; v_C)	A2	Wrong-category object: circle a clearly different object than the target.	✓
	A3	Near-miss circle: offset the centre so the target falls outside or only grazes the circle.	✓
	A4	Empty-region circle: circle bare table or background.	✓
	A5	Multi-object enclosure: inflate the radius to enclose the target and a distractor.	✓
	A6	Degenerate size: dot-sized or whole-frame circle.	✓
Directional (arrow / "where"; v_A)	B1	Wrong destination: retarget the head to a different container or region.	✓
	B3	Wrong magnitude: correct bearing, arrow far too short or overshooting.	✓
	B4	Wrong bearing: correct length, head rotated $90^\circ/180^\circ$.	✓
	B5	Null destination: head on empty table or off-frame.	✓
	B6	Detached tail: shaft plausible but tail not anchored on the circled object.	✓
	B7	Spec mismatch: arrow present on a circle-only task, or missing on a place task.	✓
Cross-modal (v_X)	C1	Circle–arrow mismatch: circle object A while the arrow originates from object B.	✓
	C2	Caption–sketch mismatch: correct sketch, caption swapped from another task.	N/A
	C3	Predicate mismatch: caption says "open the cabinet" while the sketch is a place-style circle+arrow.	✓
Language (caption / "what"; v_L)	D1	Wrong target noun.	N/A
	D2	Wrong destination noun.	N/A
	D3	Wrong verb (pick vs. push vs. open).	✓
	D4	Under-determined caption: "put this there", valid with a good sketch, invalid with a bad one.	✓
Grounding (v_S)	–	Five configurations: resolved; no sketch; open destination; twin referents; arrow between destinations.	✓

Table 2: DrawVerify v1 composition by label (train / validation / test). Percentages are shares within a split; the label balance is held constant across splits, and no demonstration contributes frames to more than one split.

Label	Train	Val	Test	Total
Valid	2,057 (31.8%)	285 (31.7%)	366 (31.6%)	2,708 (31.8%)
Invalid	4,406 (68.2%)	615 (68.3%)	793 (68.4%)	5,814 (68.2%)
Total	6,463	900	1,159	8,522

B Pilot Results, Training Details, and Protocols

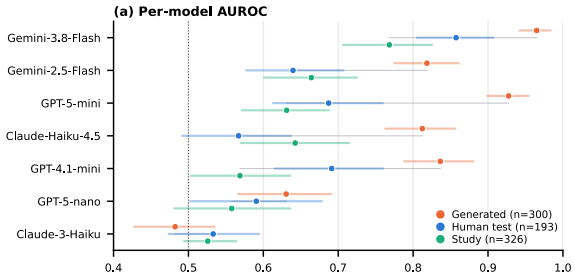


Figure 3: Pilot evidence (referenced from Section 4): AUROC of seven API validators on generated prompts, the human test split and the in-the-loop study (95% bootstrap intervals). Generated prompts rank models like human sketches do but inflate absolute scores unevenly, motivating H1 and H3.

Table 3: Pilot: seven API models under the identical 4-shot prompt on generated prompts (Gen., $n = 300$, 60.0% invalid), the human test split (Test, $n = 193$, 42.5% invalid) and the in-the-loop study (Study, $n = 326$, 23.6% invalid). The last rows give each human set's Spearman agreement with the generated ranking and the fraction of the 21 model pairs whose order is preserved.

	AUROC			MCC		
Model	Gen.	Test	Study	Gen.	Test	Study
Gemini-3.8-Flash	**0.965**	**0.857**	**0.768**	+0.85	+0.50	+0.43
GPT-5-mini	0.928	0.687	0.631	+0.82	+0.24	+0.25
GPT-4.1-mini	0.836	0.691	0.569	+0.64	+0.24	+0.15
Gemini-2.5-Flash	0.818	0.640	0.664	+0.65	+0.29	+0.25
Claude-Haiku-4.5	0.812	0.567	0.642	+0.61	+0.10	+0.27
GPT-5-nano	0.630	0.591	0.558	+0.31	+0.26	+0.10
Claude-3-Haiku	0.482	0.533	0.526	−0.03	+0.07	+0.10
Mean	0.782	0.652	0.623	+0.55	+0.24	+0.22
ρ vs. Gen.		+0.93	+0.71		+0.71	+0.75
Pairs preserved		19/21	17/21		17/21	17/21

Table 4: Pilot: fine-tuned validators against Gemini-3.8-Flash on the human test split ($n = 193$, 42.5% invalid). Positive class is INVALID; Prec and Rec are on invalid prompts, R_{val} is recall on valid ones. Δ is the paired AUROC difference to Gemini-3.8-Flash, * marking a 95% bootstrap interval excluding zero. "gen→hum" is the staged recipe; decisions are the generated verdict.

Model	Acc	Prec	Rec	MCC	R_{val}	AUROC	Δ
Gemini-3.8-Flash (4-shot)	0.699	0.590	0.963	$+0.50$	0.505	0.857	
Qwen3-VL-4B zero-shot	0.585	0.600	0.073	$+0.08$	0.964	0.542	-0.32^*
Qwen3-VL-4B generated only	0.653	0.559	0.866	$+0.38$	0.495	0.565	-0.29^*
Qwen3-VL-4B human only (82)	0.627	0.561	0.561	$+0.24$	0.676	0.640	-0.22^*
Qwen3-VL-4B mixed (pooled)	0.684	0.598	0.780	$+0.39$	0.613	0.754	-0.10^*
Qwen3-VL-4B gen→hum	0.798	0.779	0.732	$+0.58$	0.847	0.808	-0.05
answer-only	0.808	0.817	0.707	$+0.61$	0.883	0.841	-0.02
40 human sketches	0.782	0.900	0.549	$+0.57$	0.955	0.847	-0.01
Qwen3-VL-2B gen→hum	0.762	0.757	0.646	$+0.51$	0.847	0.779	-0.08^*
Qwen2-VL-2B gen→hum	0.741	0.767	0.561	$+0.46$	0.874	0.720	-0.14^*
Geometry-only LightGBM (gen. fit)	–	–	–	–	–	0.741	*

Fine-tuning (pilot recipe). LoRA rank 32, $\alpha = 64$, dropout 0.05 on the query, key, value and output projections and the gate, up and down MLP projections (0.93% of Qwen3-VL-4B's parameters). Stage 1: 2 epochs on the generated tier, effective batch 16, learning rate 10^{-4} with cosine schedule and 30 warm-up steps (AdamW, no weight decay, gradient clipping 1.0). Stage 2: 4 epochs on the 82 human training sketches, batch 8, learning rate 3×10^{-5}. Images 512×512; loss on the assistant turn only. At inference the model generates its reasoning greedily; the prefix up to `<answer>` is forced and the decision and AUROC score are read from the relative probabilities of the VALID/WRONG tokens. Seeds: 0.810 ± 0.003 AUROC for the staged 4B model.

API protocol. The same system prompt, the same four in-context examples drawn from the human training split, temperature 0, and a request for a label and a failure probability $p_{\text{fail}} \in [0, 100]$, from which AUROC is computed with invalid as the positive class. Unparsable replies are dropped (6, 4 and 20 for GPT-5-nano on the three sets; none for any other model).

Human data. Ungated corpus: 11 participants, 275 sketches over two rounds covering 34 tasks across the four LIBERO suites; median drawing time 8.2 s; each sketch labelled valid/invalid by human annotators independently of any model, invalid ones with a failure code from the taxonomy; task-disjoint split 82 train / 193 test. In-the-loop study: 17 participants, 21 trials each in three blocks (no gate, model gate, control gate rejecting on the ground-truth geometry of the original annotation); 250 analysed trials, 326 labelled sketches. Outcomes: validity of the finally sent instruction 0.841 (no gate), 0.795 (model gate, 1.38 attempts/trial), 0.726 (control gate); the model gate's in-loop precision 0.382 and recall 0.500; trials accepted at the second attempt were only repaired (6/0), trials that reached a third broke more than they repaired (6 broken / 2 repaired); of six third-attempt breaks, five changed the prompt rather than the drawing. Replaying the logs with a one-retry cap gives 0.821 (model) and 0.762 (control).

Pilot error profile. On the human test split the staged 4B validator's recall on invalid prompts by family is 0.80 (referential), 0.53 (directional) and 0.77 (cross-modal and grounding) against 0.96 / 0.95 / 1.00 for Gemini-3.8-Flash, which buys that recall by rejecting half of the valid sketches ($R_{\text{val}} = 0.505$). Directional failures are the weakest slice, which motivates H2; geometry-only fits trained on generated data reach 0.741 AUROC on the human test split but only 0.53–0.56 on study sketches and 0.50–0.54 on the ungated ones, every fit significantly below Gemini-3.8-Flash.